# THE COTANGENT COMPLEX OF A SYMMETRIC ALGEBRA PRESENTED BY A FREE MODULE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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*Allowed standard prerequisites (editorial).*

- The naive cotangent complex of a presentation and the module of differentials of the presented algebra. the author cites Algebra, Section algebra-section-netherlander at proof line 2077. Supplies the object computed by the statement.
- Description of the kernel of the induced map of symmetric powers as the image of the syzygy module tensored with the lower symmetric powers. the author cites Algebra, Lemma algebra-lemma-presentation-sym-exterior at proof lines 2087-2089. Gives the degreewise form of the kernel that the computation starts from.
- Multiplicativity of the symmetric powers, so that the square of the kernel is generated by products of its graded pieces. used at proof lines 2090-2100. Allows the square of the ideal to be described degreewise.

*Source reference.* Original lemma 07EV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9460–9460. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 00S0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1022–1022. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 00DO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 504–504. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $A$ be a ring. Let $0 \to K \to A^{\oplus m} \to M \to 0$ be a sequence of $A$-modules. Consider the $A$-algebra $C = Sym^*_A(M)$ with its presentation $\alpha : A[y_1, \ldots, y_m] \to C$ coming from the surjection $A^{\oplus m} \to M$. Then*

$$NL(\alpha) = (K \otimes_A C \to \bigoplus\nolimits_{j=1,\ldots,m} C\, dy_j)$$

*(see Algebra, Section 00S0) in particular $\Omega_{C/A} = M \otimes_A C$.*

*Proof.* Let $J = \mathrm{Ker}(\alpha)$. The lemma asserts that $J/J^2 \cong K \otimes_A C$. Note that $\alpha$ is a homomorphism of graded algebras. We will prove that in degree $d$ we have $(J/J^2)_d = K \otimes_A C_{d-1}$. Note that

$$J_d = \mathrm{Ker}(\mathrm{Sym}^d_A(A^{\oplus m}) \to \mathrm{Sym}^d_A(M)) = \mathrm{Im}(K \otimes_A \mathrm{Sym}^{d-1}_A(A^{\oplus m}) \to \mathrm{Sym}^d_A(A^{\oplus m})),$$

see Algebra, Lemma 00DO. It follows that $(J^2)_d = \sum_{a+b=d} J_a \cdot J_b$ is the image of

$$K \otimes_A K \otimes_A \mathrm{Sym}^{d-2}_A(A^{\otimes m}) \to \mathrm{Sym}^d_A(A^{\oplus m}).$$

The cokernel of the map $K \otimes_A \mathrm{Sym}^{d-2}_A(A^{\otimes m}) \to \mathrm{Sym}^{d-1}_A(A^{\oplus m})$ is $\mathrm{Sym}^{d-1}_A(M)$ by the lemma referenced above. Hence it is clear that $(J/J^2)_d = J_d/(J^2)_d$ is equal to

$$\begin{aligned} \mathrm{Coker}(K \otimes_A K \otimes_A \mathrm{Sym}^{d-2}_A(A^{\otimes m}) \to K \otimes_A \mathrm{Sym}^{d-1}_A(A^{\otimes m})) &= K \otimes_A \mathrm{Sym}^{d-1}_A(M) \\ &= K \otimes_A C_{d-1} \end{aligned}$$

as desired. □